



Regular Research Article

The skill premium across countries in the era of industrial robots and artificial intelligence[☆]Marcos J. Ribeiro^a, Klaus Prettnner^{b,*} ^a Faculdade de Economia, Administração e Contabilidade de Ribeirão Preto da Universidade de São Paulo, Avenida Bandeirantes, 3900 - Vila Monte Alegre, Ribeirão Preto, SP, 14040-905, Brazil^b Vienna University of Economics and Business, Welthandelsplatz 1, 1020 Vienna, Austria

HIGHLIGHTS

- Robots raise wage inequality, AI reduces it.
- The effects are stronger in high-income countries.
- Both robots and AI boost economic growth.
- 1% increase in robots raises GDP by 0.29% and the skill premium by 0.23%.
- 1% increase in AI raises GDP by 0.03% and reduces the skill premium by 0.018%.

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ABSTRACT

How do new technologies affect economic growth and the skill premium? To answer this question, we analyze the impacts of industrial robots and artificial intelligence (AI) across 52 countries using counterfactual simulations. In so doing, we simulate the nested CES production function framework of Bloom et al. (2025) with low-skilled workers, high-skilled workers, traditional capital, robots, and AI. Our contribution lies in documenting substantial heterogeneity across income groups: the inequality-increasing effect of robots and the inequality-reducing effect of AI are particularly strong in high-income countries, while they are less pronounced among upper-middle and lower-middle income countries. In addition, we show that both technologies boost economic growth. In terms of policy recommendations, our framework suggests i) balanced investment in both types of automation technologies to increase economic growth while keeping wage inequality subdued, and ii) investments in education and skill-upgrading to mitigate negative effects of robot use on wage inequality.

1. Introduction

Automation of production processes has risen substantially over the past decade. According to the International Federation of Robotics (IFR), the number of operative industrial robots increased by 689% worldwide between 1993 and 2023. Even more impressive, private investment in artificial intelligence (AI) increased by 1159% between 2013 and 2023.¹ AI models such as ChatGPT are already widely used to translate and write texts, generate computer code, analyze large volumes of data

quickly and accurately, diagnose diseases, make accurate predictions that are useful in materials science and drug development, and for many other purposes (cf. Deng & Lin, 2023; Prettnner & Bloom, 2020).²

While industrial robots and AI boost economic growth, they can have adverse effects on the labor market, such as job displacement and increased wage inequality (see Acemoglu & Restrepo, 2020; Brambilla et al., 2023; Dauth et al., 2021; Gregory et al., 2022; Mann & Pittmann, 2023; Stemmler, 2023, for different findings in the US, Europe, and Latin

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¹ See Fig. A.1.

² While automation involves replacing human workers with machines programmed through some type of software, AI refers to a type of software that can mimic the thinking process of humans and also has the ability to learn from data (Deng & Lin, 2023; Growiec, 2019; Prettnner & Bloom, 2020).

America). According to [Lankisch et al. \(2019\)](#), automation by means of industrial robots predominantly replaces low-skilled workers in routine tasks and raises the skill premium—the wage differential between high-skilled and low-skilled workers. By contrast, [Bloom et al. \(2025\)](#) show that the use of AI has the potential to reduce the skill premium.

The framework proposed by [Bloom et al. \(2025\)](#) relies on a nested CES production function with three distinct types of capital (traditional physical capital, industrial robots, and AI) and two types of labor (low-skilled and high-skilled). In this setting, the effects of AI and industrial robots on the skill premium depend on two main factors: the extent to which these technologies are already embedded in production and the elasticities of substitution between the different types of capital and labor. As a result, the effects are larger in countries where AI and robots already play a more important role in the production process.

We contribute to the literature by providing the first cross-country application of the framework suggested by [Bloom et al. \(2025\)](#), simulating how automation in terms of industrial robots and AI affects economic growth and the skill premium across 52 countries. The impact of AI on the skill premium is strongest in high-income countries and much weaker in lower-middle income countries. This documents substantial heterogeneity across countries depending on income levels. In addition, we are able to quantify the effects of changes in robot use and AI in terms of per capita GDP growth and in terms of inequality reduction with the numbers being useful for follow-up policy analyses.

Our simulation results show that, on average, a 1% increase in industrial robots and AI per worker corresponds to a 0.29% and 0.03% increase in GDP per worker, respectively. In terms of inequality, a 1% increase in the robot stock per worker is associated with an increase in the skill premium by 0.23% on average. The rise in the skill premium tends to be more pronounced in countries where the introduction of industrial robots replaces a greater number of low-skilled workers. As far as the use of AI is concerned, a 1% rise in AI investment is associated with a 0.018% reduction in the skill premium, especially in high-income countries.

The paper is organized as follows. In [Section 2](#), we discuss the literature on the effects of industrial robots and AI on economic growth and inequality. In [Section 3](#), we revisit the nested CES production function of [Bloom et al. \(2025\)](#) and the associated skill premium. In [Section 4](#), we present our dataset and discuss the parameter values used in our analysis. In [Section 5](#), we present the results of the baseline calibration and the results of the counterfactual exercises. Finally, in [Section 6](#), we discuss the policy implications of our findings and conclude.

2. Related literature

Closely related to our contribution are studies on the impacts of automation on productivity and economic growth. According to [Kromann et al. \(2020\)](#), an increase in the intensity of robot use by one standard deviation raises total factor productivity (TFP) by more than 6.5%. Similarly, [Graetz and Michaels \(2018\)](#) point out that a greater density of robots in the industrial sector improves labor productivity, contributing to an average increase of 0.37 percentage points in economic growth rates over time across the 17 analyzed countries. As far as the growth effects of AI are concerned, findings vary widely. [Acemoglu \(2025\)](#) calculates rather limited effects of AI on productivity (0.66% over ten years) and on per capita GDP (about 1% over ten years), due to the fact that AI is not yet used productively in many tasks. Using a similar approach as [Acemoglu \(2025\)](#) but acknowledging the heterogeneity across tasks, [Filippucci et al. \(2026\)](#) arrive at a much higher estimate of a 0.9 percentage point productivity increase per year, while [Goldman Sachs \(2023\)](#) even estimates a 1.5 percentage points increase per year.³

³ There are also predictions of much faster growth accelerations in the case of the emergence of “superhuman” artificial general intelligence, which outperforms humans in all relevant tasks by a large extent (see, for example [Korinek &](#)

As far as wage inequality is concerned, [Kunst et al. \(2022\)](#) identify a U-shaped pattern in the skill premium in several countries, with a decrease between 1950 and 1980, followed by an increase from the 2000s onwards. This dynamic is initially explained by the growth in the supply of educated workers, and later by factors such as the weakening of labor unions and the increase in the demand for skilled labor, especially as a result of the expansion of trade, skill-biased technological change, and the accumulation of capital that is more complementary to high-skilled workers than to low-skilled ones ([Acemoglu, 2002](#); [Fadinger & Mayr, 2014](#); [Krusell et al., 2000](#)). Additionally, [He \(2012\)](#) shows that investment-specific technological change (ISTC) is one of the main drivers of changes in the skill premium in the US economy.⁴

In terms of middle-income economies in the context of automation, [Brambilla et al. \(2023\)](#) analyze the effects of robot adoption on local labor markets in Argentina, Brazil, and Mexico. They find a causal effect of robot adoption on job losses, particularly among young and semi-skilled workers suggesting an inequality enhancing effect. [Stemmler \(2023\)](#), in turn, finds that robot adoption in Brazil increases the ratio of female and skilled workers as compared to other parts of the workforce. This skill-complementarity also suggests inequality-enhancing effects of robot use. In addition, [Stemmler \(2023\)](#) identifies employment-reducing effects when trading partners use industrial robots. This suggests a role for automation-driven re-shoring tendencies in the economies of the trading partners of middle-income countries (see also [Krenz et al., 2021](#)).

Focusing on AI, [Noy and Zhang \(2023\)](#) provide experimental evidence across 453 college-educated professionals that access to ChatGPT leads to a productivity boost that is more pronounced among low-skilled workers than among high-skilled workers. Similarly, [Brynjolfsson et al. \(2025\)](#) find that access to an AI assistant increased productivity among 5172 customer-support agents with low-skilled workers benefiting more from AI than high-skilled workers. In addition, [Autor \(2024\)](#) provides a very optimistic outlook for the inequality-reducing effects of AI suggesting that it could help in rebuilding the middle class because tasks that were previously out of its reach can now be performed by them with the help of AI. This perspective emphasizes the labor-augmenting nature of AI instead of its labor-displacing one. Somewhat in contrast to the previously discussed literature, [Dell’Acqua et al. \(2026\)](#) find that the heterogeneity of tasks matters (see also [Acemoglu & Autor, 2012](#); [Autor, 2013](#), for the description of a task-based framework to the labor market). According to their results, in complex tasks, the use of AI may actually reduce productivity.

With respect to the disparities in the skill premium across countries with different levels of development, [Banerjee et al. \(2023\)](#) show that high-skilled individuals tend to earn relatively more in poorer countries than in richer ones. They estimate that a 1% increase in GDP per capita has two distinct effects: a 0.16 percentage point increase in the share of high-skilled workers and a 0.54 percentage point reduction in the wage premium for these professionals. These results suggest that as an economy develops, the supply of high-skilled workers increases, leading to a reduction in the skill premium.

Broadly consistent with previous findings but adding the dimension of AI, we show that industrial robots and AI both increase per capita GDP. With respect to wage inequality, however, the two technologies have differential effects. Industrial robots tend to amplify wage inequality, while AI tends to lessen it. In addition, the inequality-related effects are more pronounced in high-income countries than in low-income countries. Finally, our simulations imply that investing in skill-upgrading could reduce the adverse effects of robot use on wage inequality.

[Suh, 2024](#); [Trammell & Korinek, 2023](#)). However, this is a hypothetical future scenario that we do not explore here.

⁴ Investment-specific technological change (ISTC) refers to advancements in technology that specifically enhance the efficiency and productivity of capital goods ([Greenwood et al., 1997](#)).

3. The model

Bloom et al. (2025) propose the following nested constant elasticity of substitution (CES) production function to analyze the effects of AI on economic outcomes, particularly on the skill premium:

$$Y_t = A_t K_t^\alpha \left[\beta_3 \left(\beta_1 (L_{u,t})^\theta + (1 - \beta_1) P_t^\theta \right)^{\frac{\gamma}{\theta}} + (1 - \beta_3) \left(\beta_2 (L_{s,t})^\varphi + (1 - \beta_2) G_t^\varphi \right)^{\frac{\gamma}{\varphi}} \right]^{\frac{1-\alpha}{\gamma}} \quad (1)$$

In this function, Y_t represents the economy's output in period t , K_t corresponds to traditional physical capital (machines, assembly lines, etc.), P_t refers to the stock of industrial robots, and G_t to the stock of AI. In addition, $L_{s,t}$ and $L_{u,t}$ denote the quantities of high-skilled and low-skilled workers, respectively, and the term A_t represents TFP, which is added to the function for calibration purposes.

The remaining parameters are constant, with α being the elasticity of output with respect to physical capital, θ determining the degree of substitutability between low-skilled labor and industrial robots, φ determining the degree of substitutability between high-skilled labor and AI, and γ determining the degree of substitutability between the combination of low-skilled labor and robots, and the combination of high-skilled labor and AI. Finally, β_1 , β_2 , and β_3 represent the weights attributed to low-skilled labor, high-skilled labor, and the combination of low-skilled labor with robots in the production process, respectively.

As in Bloom et al. (2025), the ratio between the marginal products of high-skilled workers and low-skilled workers yields the skill premium

$$\frac{w_s}{w_u} = \frac{\beta_2(1 - \beta_3)}{\beta_1\beta_3} \frac{\left[\beta_2 L_{s,t}^\varphi + (1 - \beta_2) G_t^\varphi \right]^{\frac{\gamma-\varphi}{\varphi}} \left(\frac{L_{s,t}^{\varphi-1}}{L_{u,t}^{\varphi-1}} \right)}{\left[\beta_1 L_{u,t}^\theta + (1 - \beta_1) P_t^\theta \right]^{\frac{\gamma-\theta}{\theta}} \left(\frac{L_{s,t}^{\theta-1}}{L_{u,t}^{\theta-1}} \right)} \quad (2)$$

The closer this measure is to unity, the more equal the wages are across the two skill groups. From this expression, Bloom et al. (2025) derive the following theoretical effects that we analyze in counterfactual simulations across countries:

- **Effect of AI on the skill premium:** The increasing use of AI, *ceteris paribus*, reduces the wage gap between high-skilled and low-skilled workers as long as the degree of substitutability between high-skilled labor and AI (φ) is greater than the degree of substitutability between low-skilled and high-skilled intensive production tasks (γ).
- **Effect of industrial robots on the skill premium:** The increasing use of industrial robots, *ceteris paribus*, increases the wage gap between high-skilled and low-skilled workers, as long as the degree of substitutability between low-skilled labor and industrial robots (θ) is greater than the degree of substitutability between low- and high-skilled intensive production tasks (γ).

To see this, we compute the elasticity of the skill premium with respect to AI, defined as the logarithmic derivative of the skill premium with respect to AI:

$$\frac{\partial \ln(w_s/w_u)}{\partial \ln G_t} = (\gamma - \varphi) S_{G_t}, \quad (3)$$

where

$$S_{G_t} = \frac{(1 - \beta_2) G_t^\varphi}{\beta_2 L_{s,t}^\varphi + (1 - \beta_2) G_t^\varphi} \quad (4)$$

captures the relative importance of AI in the high-skill-labor composite. Assuming $0 < \beta_2 < 1$ and that all variables and remaining parameters are strictly positive for all $t \geq 0$, the sign of Eq. (3) depends only on the term $(\gamma - \varphi)$. Therefore, the skill premium w_s/w_u is decreasing in G_t if and only if $\varphi > \gamma$ (Bloom et al., 2025).

Second, we compute the logarithmic derivative of the skill premium with respect to the use of industrial robots as

$$\frac{\partial \ln(w_s/w_u)}{\partial \ln P_t} = (\theta - \gamma) S_{P_t}, \quad (5)$$

where

$$S_{P_t} = \frac{(1 - \beta_1) P_t^\theta}{\beta_1 L_{u,t}^\theta + (1 - \beta_1) P_t^\theta} \quad (6)$$

captures the relative importance of robots in the low-skill-labor composite. Assuming $0 < \beta_1 < 1$, and that all variables and remaining parameters are strictly positive for all $t \geq 0$, the sign of Eq. (5) depends only on the term $(\theta - \gamma)$. Therefore, the skill premium w_s/w_u increases in P_t if and only if $\theta > \gamma$.

Additionally, the relative importance of AI in the high-skill-labor composite, S_{G_t} , and the relative importance of robots in the low-skill-labor composite, S_{P_t} , scale the effects of AI and robots on the skill premium. In countries where these technology shares are higher, an increase in AI or robots has a larger effect on the skill premium than in countries where these shares are lower.

Finally, from Eq. (1), it is clear that increases in robot use (higher P_t) and increases in AI use (higher G_t) will both raise income per capita and labor productivity as measured by output per worker. The extent to which these increases materialize depends on the underlying country characteristics and the magnitude of the changes in P_t and G_t . From these observations the following remark can be stated.

Remark 1. The effects of robot adoption and AI on economic growth and wage inequality depend on the country characteristics under consideration, in particular, on the income levels and the starting values for the stock of robots and the use of AI.

Next, we use the information from different countries to assess the extent to which per capita income, productivity, and the skill premium change across countries when robot use and AI use change. Please note that our results do not represent empirical estimates but are simulation results of the theoretical model for the given parameter settings.

4. Dataset and parameters

In this section, we present the dataset and the calibration strategy used to simulate the cross-country skill premium.

4.1. Dataset

To examine the effects of industrial robots and AI on economic growth and the skill premium, we use Eqs. (1) and (2) in a cross-country analysis. Our sample is composed of data from 2019 and covers 52 countries for which data were available at that time. We chose 2019 because it is the most recent year for which all variables are available.⁵ According to the World Bank income classification (World Bank, 2018), 34 of the countries are considered high-income, 12 are upper-middle income, and 6 are lower-middle income.⁶ The reason why we do not include low-income countries is that industrial robot use and AI investment tend to be very low in these countries and there is a lack of reliable and precise data. The country names and codes of our sample are displayed in Table B1 of Appendix B.

⁵ The capital variable K_t , provided by the International Monetary Fund (2025), is only available until 2019.

⁶ The World Bank (2018) income classification divides countries according to their Gross National Income (GNI) per capita. Low-income countries have GNIs below US\$996, lower-middle income countries have GNIs between US\$996 and US\$3895, upper-middle income countries have GNIs between US\$3896 and US\$12,055, and high-income countries have GNIs above that amount. It should be noted that there are no low-income countries in our sample. See: <https://blogs.worldbank.org>.

Regarding the main variables used in this study and their sources, the traditional physical capital stock K was obtained from the [International Monetary Fund \(2025\)](#) database. The variables related to low-skilled and high-skilled labor, L_u and L_s , were obtained from the [International Labour Organization \(2025\)](#). In [Table A.1 of Appendix A](#), we provide the ILO methodology for classifying workers.⁷

Regarding automation technologies, the stock of industrial robots P was obtained from the [International Federation of Robotics \(2023\)](#). Since the IFR does not provide a database with average robot prices for the sample, we adopted the estimate from [Jurkat et al. \(2022\)](#) for the United States in 2019, which amounts to US\$ 52,810. To adjust robot prices internationally, we rely on the price level ratio (PLR), defined as the ratio between the purchasing power parity (PPP) conversion factor and a country's market exchange rate relative to the United States. The PLR indicates how many dollars are required in a given country to purchase the same volume of goods that US\$ 1 buys in the United States. Accordingly, the calculation $\text{PLR} \times \text{industrial robot price}$ adjusts the robot's base price (US\$ 52,810 in the United States, where $\text{PLR} = 1$) to the price in dollars in other countries, while preserving the same real value in terms of purchasing power.⁸

For the variable associated with AI, G , we used investment in AI technologies as a proxy, with data from the Center for Security and Emerging Technology and the U.S. Bureau of Labor Statistics, processed by the platform Our World in Data ([Roser, 2025](#)). We acknowledge that investment in AI is an imperfect proxy for the AI stock. Therefore, in [Section 5.3.2](#), we conduct a robustness analysis using two alternative proxies for the stock of AI. The first proxy follows [Bloom et al. \(2025\)](#) and assumes that the stock of AI is a function of the value of the stock of robots. The second proxy is constructed using the logic of the perpetual inventory method under different assumptions about the depreciation and growth rates of AI investment. In both cases, our main conclusions remain unchanged.

4.2. Calibration strategy

We calibrate TFP for the 52 countries in the sample by selecting values for A such that the GDP implied by the model coincides with the values in the data. To this end, we define the following objective function for N countries:

$$D = \sum_{i=1}^N \left(\frac{GDP_i^M - GDP_i^T}{GDP_i^T} \right)^2. \quad (7)$$

The superscripts M and T indicate the model and target statistics. The model fits well with the empirical data, resulting in a D close to zero.

Regarding the parameters of Eq. (1), we set them based on external sources and in a way that is consistent with general evidence on the substitutability across skill-specific tasks. The parameters β_1 , β_2 , and β_3 are assigned the values 0.9, 0.95, and 0.66, respectively, following [Bloom et al. \(2025\)](#). The elasticity parameters θ , γ , φ , and α are set according to the literature. Specifically, θ is set to 3/4, based on [Jurkat et al. \(2022\)](#), while α and γ are both set to 1/3, following [Acemoglu \(2002, 2009\)](#). Finally, φ is set to 1/2, satisfying the condition $0 < \varphi < \theta \leq 1$, as in [Bloom et al. \(2025\)](#).

Our parameter ranges regarding the elasticities imply that industrial robots are better substitutes for low-skilled workers than AI is for high-skilled workers. This assumption is highly plausible because industrial

Table 1
Parameter values and data sources.

Parameter	Value	Parameter type	Source
P	–	Data	International Federation of Robotics; Jurkat et al. (2022)
G	–	Data	Center for Security and Emerging Technology; U.S. BLS – processed by Our World in Data
K	–	Data	International Monetary Fund
L_u	–	Data	International Labour Organization
L_s	–	Data	International Labour Organization
A	–	Calibrated–Matched	Calibrated using Eq. (7)
θ	0.75	Calibrated–External	Jurkat et al. (2022)
α	0.33	Calibrated–External	Acemoglu (2009)
γ	0.33	Calibrated–External	Acemoglu (2002)
φ	0.5	Chosen	Chosen such that $0 < \varphi < \theta \leq 1$
β_1	0.9	Calibrated–External	Bloom et al. (2025)
β_2	0.95	Calibrated–External	Bloom et al. (2025)
β_3	0.66	Calibrated–External	Bloom et al. (2025)

robots are already widely used along assembly lines without the need for any human labor input in the tasks that they perform (which implies that the elasticity of substitution is rather high, as also estimated by [Jurkat et al., 2022](#)). By contrast, tasks for which AI is used intensively include programming, where human intervention and oversight are still needed as of now; diagnosing diseases, where it is required that a medical doctor communicates with the patient in the end; and materials science and drug development, where it is still the human scientists who interpret the results. In addition, our parameters also imply that it is easier to perform the tasks of high-skilled workers using AI than employing low-skilled workers. Again, this is plausible because the tasks of high-skilled workers that are performed by AI such as translating texts, writing computer code, diagnosing diseases, analyzing large amounts of data, etc. are very difficult to perform by low-skilled workers.

Overall, our results represent the average theoretical implications of the use of industrial robots and AI under common parameters for all countries. In reality, of course, parameters may vary across the countries under consideration. The sources of variables and parameter values are described in [Table 1](#).

5. Results

In this section, we investigate the relationship between industrial robots and AI on the one hand, and economic growth and the skill premium on the other, in 52 countries categorized by income level. In [Section 5.1](#), we present descriptive statistics of the skill premium and explore the relationship between this measure and variables such as the stock of industrial robots, investment in AI, GDP, the share of high-skilled workers, and TFP.

In [Section 5.2](#), we present the counterfactual exercises. To conduct these analyses, we consider output per worker. Dividing Eq. (1) by L and rearranging terms, we obtain

$$\begin{aligned} \frac{Y}{L} = A \left(\frac{K}{L} \right)^\alpha & \left[(1 - \beta_3) \left(\left(\frac{G}{L} \right)^\varphi (1 - \beta_2) + \left(\frac{L_s}{L} \right)^\varphi \beta_2 \right)^{\frac{\gamma}{\varphi}} \right. \\ & \left. + \beta_3 \left(\left(\frac{P}{L} \right)^\theta (1 - \beta_1) + \left(\frac{L_u}{L} \right)^\theta \beta_1 \right)^{\frac{\gamma}{\theta}} \right]^{\frac{1-\alpha}{\gamma}}. \end{aligned} \quad (8)$$

Table 2
Descriptive statistics of skill premium - 2019.

Average	Minimum	Maximum	Median	Std. Dev.
1.44	0.68	4.28	1.35	0.667

⁷ We consider medium-skilled workers as low-skilled workers for the sake of tractability. However, the results are robust to the exclusion of medium-skilled workers.

⁸ The PLR we use refers to a general basket of goods and was obtained from the World Bank database. In our context, the best approach would be to use the PLR for machinery and equipment. While the OECD database contains PPP data for machinery and equipment, which would allow us to calculate the PLR, these data cover only the period from 2020 to 2024 and are not available for all of the countries we are studying.

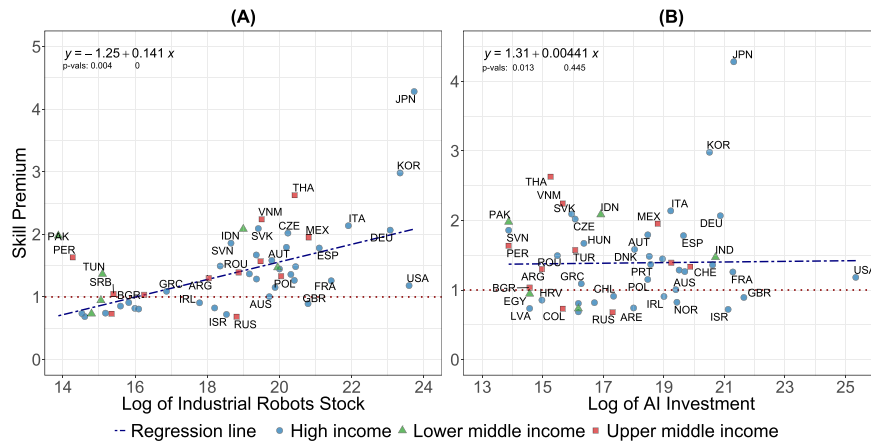


Fig. 1. Skill premium, stock of robots, and AI investments. Note: To estimate regressions robust to outliers, we employ the Huber regressor.

We can rewrite this expression in a more compact form as

$$y_t = A_t k_t^\alpha \left[\beta_3 \left(l_{u,t}^\theta \beta_1 + (1 - \beta_1) p_t^\theta \right)^{\frac{\gamma}{\theta}} + (1 - \beta_3) \left(l_{s,t}^\theta \beta_2 + (1 - \beta_2) g_t^\theta \right)^{\frac{\gamma}{\theta}} \right]^{\frac{1-\alpha}{\gamma}}, \quad (9)$$

where the terms k_t , $l_{s,t}$, $l_{u,t}$, p_t , and g_t represent, respectively, capital, high-skilled labor, low-skilled labor, industrial robots, and AI, each measured per worker. This adjustment ensures consistency with the CES production function and facilitates comparison across economies. Finally, we can also rewrite the skill premium in per worker terms⁹:

$$\frac{w_s}{w_u} = \frac{\beta_2(1 - \beta_3)}{\beta_1\beta_3} \frac{\left[\beta_2 l_{s,t}^\theta + (1 - \beta_2) g_t^\theta \right]^{\frac{\gamma-\theta}{\theta}}}{\left[\beta_1 l_{u,t}^\theta + (1 - \beta_1) p_t^\theta \right]^{\frac{\gamma-\theta}{\theta}}} \left(\frac{l_{s,t}^{\theta-1}}{l_{u,t}^{\theta-1}} \right). \quad (10)$$

In Section 5.2.1, we perform counterfactual analyses in which we equalize, in different scenarios, the composition of the labor force, the stock of robots per worker, and investment in AI per worker to the “best-practice” country with the highest level of the corresponding variables. We then compare GDP per worker and the skill premium before and after the counterfactual. In the case of the fraction of high-skilled workers, the counterfactual refers to Sweden, in the case of the stock of robots per worker, the counterfactual is given by South Korea, and for investment in AI per worker, the counterfactual is represented by the United States. Furthermore, in Section 5.2.2, we simulate gradual increases in the number of robots and in AI per worker, separately and jointly, to analyze their effects on the skill premium across countries with different income levels. Finally, in Section 5.3, we conduct robustness checks in which we vary the model parameters θ , φ , β_1 , and β_2 , and calculate the variations in the skill premium across scenarios with an increase in robots and AI.

5.1. Baseline calibration

The estimated skill premium for the 52 countries in our sample can be seen in Table B1 of Appendix B.¹⁰ In Table 2, we present descriptive statistics of these estimates. As we can see, the average skill premium is 1.44, close to that observed in Belgium (BEL). The lowest estimated

value is 0.68, close to those observed in Russia (RUS) and Iceland (ISL), while the highest value, 4.28, is observed in Japan (JPN). These differences may be related to the level of industrial robots and AI investment in each country. Japan, for example, has a large number of industrial robots and substantial investment in AI, while Iceland has a relatively limited amount of these technologies.

In Fig. 1, we present the relationship between the skill premium (vertical axis) and two variables: industrial robots (Fig. 1 (A)) and AI (Fig. 1 (B)). The countries in the sample were categorized by income level: high-income countries (blue circles), upper-middle income countries (orange squares), and lower-middle income countries (purple triangles). The horizontal line indicates where the skill premium equals one. Countries above this line are those where high-skilled workers earn more than low-skilled workers, while the reverse holds true for countries below it. We also added a regression line to indicate the general trend of the variables. Fig. 1 (A) suggests a positive correlation, indicating that countries with a greater stock of robots tend to have a higher skill premium, especially in high-income economies. This suggests that the number of industrial robots present in the production process may play an important role in the relative valuation of skills in the labor market. By contrast, in Fig. 1 (B), we notice that the slope of the regression line between the skill premium and AI investment is insignificant.

In Fig. 2, we present the relationship between the skill premium (vertical axis) and three variables: the logarithm of GDP (Fig. 2 (A)), the share of high-skilled labor (Fig. 2 (B)), and the logarithm of TFP (Fig. 2 (C)). The dashed horizontal lines refer to the average of the skill premium, while the dashed vertical lines represent the average of the logarithm of GDP (Figure A), the average of the share of high-skilled labor (Figure B), and the average of TFP (Figure C).

Fig. 2 (A) indicates a positive relationship between the skill premium and GDP, suggesting that economies with higher GDP tend to have a larger wage gap between high- and low-skilled workers. Regarding the level of income, high-income countries such as Japan (JPN) and South Korea (KOR) have a high GDP and a high skill premium. Upper-middle income and lower-middle income countries, for example, Thailand (THA), Vietnam (VNM), Mexico (MEX), and Indonesia (IDN) show a similar pattern.

In Fig. 2 (B), the relationship between the share of high-skilled workers and the skill premium is negative. High-income countries generally have a greater share of high-skilled workers and a lower skill premium, while middle-income economies show greater variability. According to a standard labor market model, this effect is expected, as a greater supply of skilled workers reduces the relative shortage of these professionals, putting downward pressure on their wages and the skill premium. However, the effect is not in line with the notion of endogenous skill-biased technological change, according to which the skill premium rises

⁹ The skill premium in Eq. (10) is identical to that in Eq. (2). We choose to rewrite it in per worker terms just to simplify the implementation of the counterfactual exercises.

¹⁰ In Appendix B, we also compare the model-generated skill premium with the skill premium computed from the data and show that the two measures are positively associated.

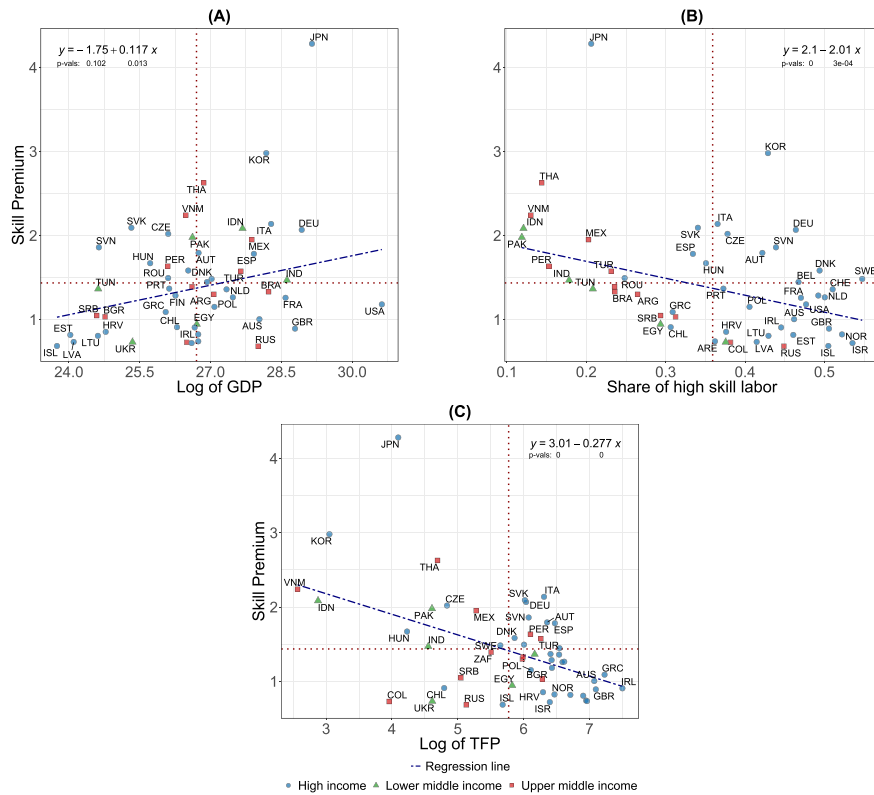


Fig. 2. Skill Premium, GDP, share of high-skilled workers, and TFP. *Notes:* (1) the dotted horizontal lines in graphs (A) and (B) refer to the average of the skill premium. The dotted vertical line represents the average of the logarithm of GDP and the share of high-skilled labor, respectively. (2) the average skill premium is 1.44. (3) to estimate regressions robust to outliers, we employ the Huber estimator.

further when the supply of high-skilled workers increases (Acemoglu, 2002).

Fig. 2 (C) reveals a negative correlation between the skill premium and TFP, suggesting that countries with higher productivity tend to have a smaller wage gap between high-skilled and low-skilled workers. This finding may be associated with the fact that countries with high productivity generally have better-developed educational systems, resulting in a greater proportion of the population with higher education or advanced technical training. This pattern can be observed in Fig. 2 (B), which shows that most high-income countries have an above-average share of high-skilled workers. Consequently, as the supply of high-skilled workers increases, the wage premium for these professionals tends to decrease.

In Fig. 3, we present the relationship between the logarithm of GDP generated by the model (horizontal axis) and two variables: the logarithm of the stock of robots (Fig. 3 (A)) and the logarithm of investment in AI (Fig. 3 (B)). In Fig. 3 (A), we observe a positive relationship between GDP and the stock of robots, as expected. Countries above the blue line have more robots than would be expected based on their GDP (for example, Germany (DEU), Italy (ITA), Japan (JPN), and South Korea (KOR)), and countries below the line have fewer robots than the size of their economies would predict (for example, the United Kingdom (GBR), Norway (NOR), and the United States (USA)). Overall, high-income countries are predominantly above the observed average of robots, which is approximately 20. Japan (JPN) and South Korea (KOR) stand out as the countries with the largest stocks of robots, while Iceland (ISL) has a low stock of robots and a low absolute level of GDP because of its small size. There are also upper-middle income countries and lower-middle income countries that have above-average robot stocks, for example, Mexico (MEX), Russia (RUS), Thailand (THA), Indonesia (IDN), and India (IND).

In Fig. 3 (B), we observe a positive relationship between GDP and AI investment. High-income countries lead in AI investment, with the United States standing out with values well above the average. The United States also has much higher investment levels in AI than predicted by the size of its economy. Some upper-middle income and lower-middle income countries have above-average AI investment, such as Brazil (BRA), Mexico (MEX), India (IND), and South Africa (ZAF). By contrast, some other upper-middle income and lower-middle income countries, such as Peru (PER), Bulgaria (BGR), and Pakistan (PAK), have substantially lower investment, falling below the average on both axes.

5.2. Counterfactual simulations

In this section we present our counterfactual simulations, which explore the theoretical predictions assuming common structural parameters across countries. Please note that the country differences we find could also potentially reflect different underlying elasticities across countries that our framework does not capture.

5.2.1. Labor force, robots, and AI

The first counterfactual analysis we conducted is on the impacts of equalizing the workforce composition of different countries (the share of high-skilled workers) to that of Sweden (SWE), the country with the greatest share of high-skilled workers. The results of this experiment are illustrated in Fig. 4. Fig. 4 (A) shows GDP per worker before and after the change in workforce composition, while Fig. 4 (B) compares the skill premium between high- and low-skilled workers before and after the adjustment.

From Fig. 4 (A), we note that all countries analyzed show little or no increase in GDP per worker after the adjustment. This is evidenced by the concentration of points above, but very close to, the 45° line,

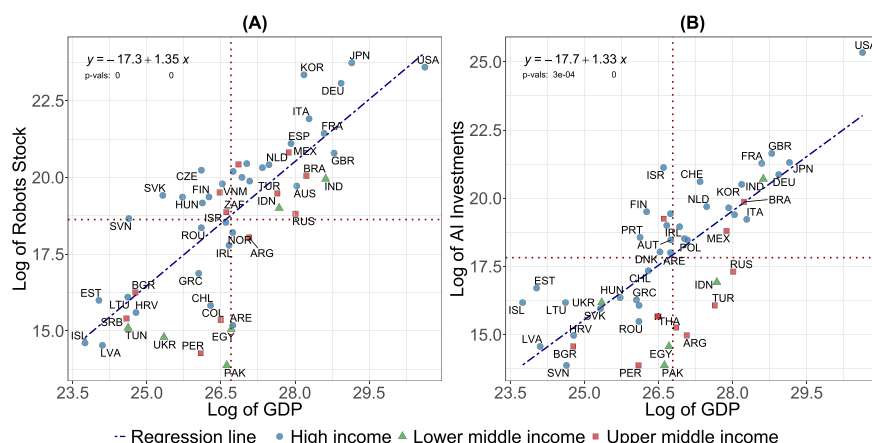


Fig. 3. Industrial Robots, AI investments, and GDP. *Notes:* (1) The dotted horizontal lines in graphs (A) and (B) refer to the averages of the logarithm of the stock of robots and investment in AI, respectively, while the dotted vertical lines represent the average of the logarithm of GDP. (2) To estimate regressions robust to outliers, we employ the Huber regressor.

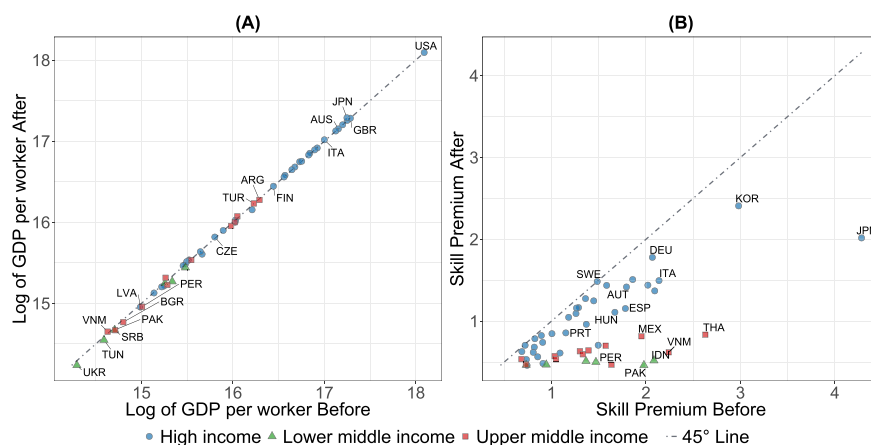


Fig. 4. GDP per worker and the skill premium before and after assigning Sweden's share of high-skilled workers to other countries. *Notes:* (1) In this counterfactual exercise, we insert into the other countries the share of high-skilled workers observed in Sweden. (2) Points above (below) the 45° line indicate that the counterfactual exercise increased (decreased) GDP per worker and the skill premium.

suggesting that countries would benefit marginally from adopting a workforce structure similar to Sweden's. On average, a 1% increase in the share of high-skilled workers increases GDP per worker by 0.001%.

By contrast, Fig. 4 (B) shows that all points fall below the 45° line, indicating a reduction in the skill premium. On average, a 1% increase in the share of high-skilled workers is associated with a decrease in the skill premium by 0.71%. This result suggests that increasing the proportion of high-skilled workers may contribute to reducing wage inequality in the analyzed economies (see also Gallego, 2012).

In the next counterfactual exercise, we simulate the impact of equalizing the stock of robots per worker in all countries to the stock observed in South Korea (KOR), the country with the greatest stock of robots per worker in 2019. In Fig. 5 (A), we compare GDP per worker of the different countries before and after the change in the robot stock. We observe that all of the points are located above the 45° line, indicating that all countries experience a rise in output per worker with a robot stock per worker similar to that of South Korea. On average, a 1% increase in the robot stock per worker is associated with an increase in GDP per worker by 0.29%.

Less developed countries, with a lower stock of robots per worker, such as Pakistan (PAK), Peru (PER), Egypt (EGY), and Colombia (COL), benefit the most from an increase in robot use. At the other end of the spectrum, countries closer to the 45° line, such as Japan (JPN) and Germany (DEU), have stocks of robots per worker more similar to South

Korea, so the positive effects of increasing the robot stock further tend to be smaller.

In Fig. 5 (B), we compare the skill premium before and after the increase in the robot stock. We observe that all points are located above the 45° line, meaning that there is a substantial increase in the skill premium. On average, a 1% increase in the robot stock per worker is associated with an increase in the skill premium by 0.23%. Countries with the largest changes in the skill premium tend to be those farthest from the technological frontier, where the introduction of robots displaces more low-skilled workers, widening the wage gap between high- and low-skilled workers.

In the next counterfactual exercise, we simulate the impact of matching the investment in AI per worker of all countries to that observed in the United States, the country with the highest investment in AI per worker. In Fig. 6 (A), where we compare GDP per worker before and after the counterfactual, we observe that all countries are above the 45° line, indicating that if other countries had the same level of investment in AI per worker as the United States, they would experience a rise in output. On average, a 1% increase in AI investment is associated with a 0.03% increase in GDP per worker. This suggests that AI adoption can increase productivity, especially in countries with low levels of AI investment.

In Fig. 6 (B), where we compare the skill premium before and after the counterfactual, we notice that all points are below the 45° line,

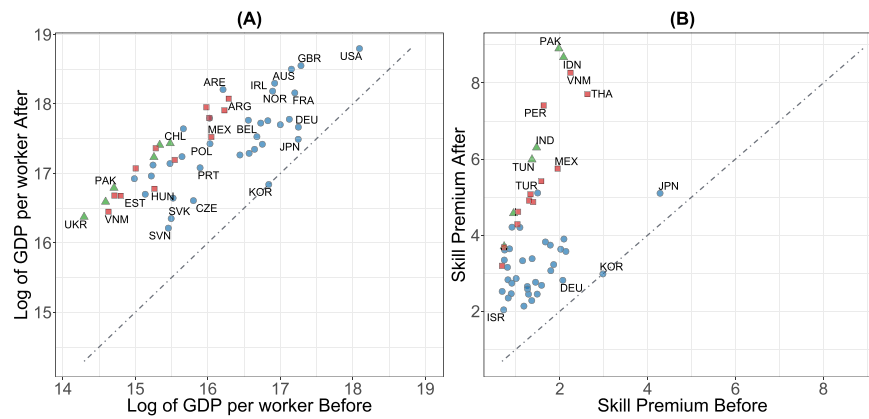


Fig. 5. GDP per worker and the skill premium before and after assigning South Korea's stock of industrial robots per worker to other countries. *Notes:* (1) In this counterfactual exercise, we insert into the other countries the robot stock per worker as observed in South Korea. (2) Points above (below) the 45-degree line indicate that the counterfactual exercise increased (decreased) GDP per worker and the skill premium.

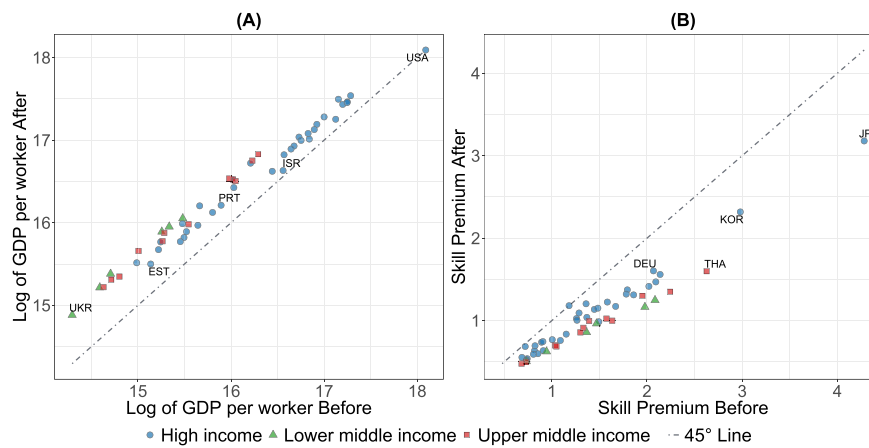


Fig. 6. GDP per worker and the skill premium before and after assigning the U.S. value of AI investment per worker to other countries. *Notes:* (1) In this counterfactual exercise, we insert into the other countries the AI investment per worker as observed in the United States. (2) Points above (below) the 45° line indicate that the counterfactual exercise increased (decreased) GDP and the skill premium.

indicating a reduction in the skill premium after matching AI investment. Quantitatively, each 1% increase in AI investment is associated with a fall in the skill premium by 0.018% on average. This suggests that AI adoption has a substitution effect, reducing the relative demand for high-skilled workers in some economies, which, in turn, induces lower wage inequality among workers.

In a pessimistic scenario, in which all countries exhibit the same level of robots per worker as Pakistan (the country with the lowest adoption of this technology) and the same level of AI investment per worker as Tunisia (the country with the lowest relative AI investment), the behavior of GDP per worker and the skill premium would be different. In the first case, a 1% reduction in the number of robots per worker would be associated, on average, with a 0.56% decrease in GDP per worker and a 0.44% decrease in the skill premium. In the second case, a 1% reduction in AI investment would be associated with a 0.12% decrease in GDP per worker and a 0.09% increase in the skill premium.

In summary, our results show that assigning South Korea's level of robots per worker to all countries is associated with an increase in the skill premium across the entire sample. Conversely, assigning the level of AI investment per worker of the United States is associated with a reduction in the skill premium for all countries. In a pessimistic scenario in which robot use and AI investment decrease, the opposite results would hold for the skill premium. Finally, in a median scenario in which every country is assigned the median levels of robots per worker and AI

investment per worker, the skill premium is higher in 50% of countries and lower in 50% of countries in both exercises.¹¹

5.2.2. Joint increases in robots and AI per worker

In this subsection, we conduct a counterfactual exercise to evaluate how changes in the stock of robots and investment in AI jointly relate to the skill premium. First, we assume a tenfold increase in both factors. Analyzing the data, we observe that, globally, the value of industrial robots increased by approximately 700% between 1993 and 2023, while investment in AI grew by more than 1000% between 2013 and 2023.¹² Given this scenario, considering a tenfold increase in production factors allows us to assess the behavior of the skill premium in the longer term. To implement this exercise, we construct a grid ranging from 1 to 10, apply it to scale the values of robots and AI per worker, and compute the skill premium at each point. We then repeat the procedure for each group of countries by income level and compute the corresponding average skill premium. The results are presented in Fig. 7.

We find that, holding the share of AI constant, an increase in the share of industrial robots is associated with an increase in the skill premium. In high-income countries (blue), the increase in the skill premium is steep. In upper-middle income countries (green), the skill

¹¹ See Fig. C1 in Appendix C.

¹² See Fig. A.1 in Appendix A.

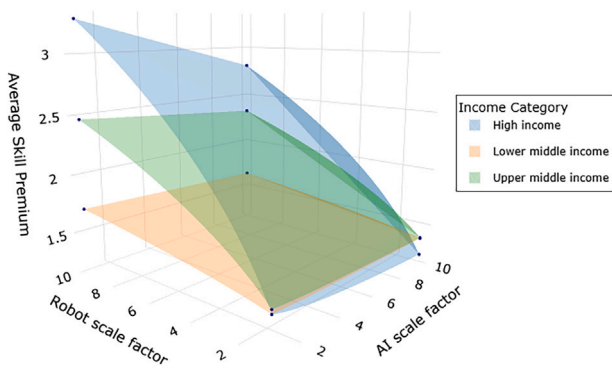


Fig. 7. Increases in industrial robots and AI per worker and their effects on the average skill premium by income level. *Note:* The navy blue dots indicate the vertices of the planes. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

premium also increases, but less steeply. For lower-middle income countries (beige), the skill premium is relatively stable, with little variation as the share of robots increases. Thus, the use of industrial robots per worker benefits high-skilled workers in high-income countries the most.

Regarding AI, we note that the opposite holds true when the share of industrial robots is constant. In high-income countries (blue), the increase in the share of AI per worker is associated with a sharp reduction in the skill premium, suggesting that AI may be replacing high-skill tasks particularly strongly in this group of countries. In upper-middle income countries (green), the relation between AI and the skill premium is weaker, but there is still a downward trend. In lower-middle income countries (beige), the relation is nearly flat. The adoption of AI appears to have a substitution effect in high-income economies, potentially displacing high-skilled workers from some roles. In middle-income economies, this effect is more muted, possibly due to differences in the labor market structure.

We also observe that the combined increase in the share of robots and AI per worker in the production process is associated with an increase in the skill premium, driven by the higher number of robots per worker. However, this effect is mitigated by the increased share of AI per worker. This suggests that, while automation through robots may widen wage inequality by increasing the skill premium, the introduction of AI may act as a compensating factor, as predicted by Bloom et al. (2025).

Still in Fig. 7, we note that the effects of robots and AI on the skill premium are larger in high-income countries. From Eqs. (3) and (5), we can see that the effects of AI and robots on the skill premium are governed by the relative importance of robots in the low-skill-labor composite, S_P , and the relative importance of AI in the high-skill-labor composite, S_G , together with the corresponding elasticity parameters. Since we assume that all countries share the same set of parameters, cross-country differences in the counterfactual responses arise from differences in these nest-specific technology shares.

Fig. 8 shows the relationship between the two technology-share measures. The vertical dashed line represents the sample average of S_G , while the horizontal dashed line represents the sample average of S_P . The figure indicates a positive association between the relative importance of AI in the high-skill-labor composite and the relative importance of robots in the low-skill-labor composite. Developed economies, such as the United States, Japan, and South Korea, tend to display relatively high values for both measures. In contrast, less developed economies, such as Pakistan, Egypt, and Peru, tend to be concentrated in the lower-left region of the figure, indicating lower relative importance of both technologies in their corresponding CES nests. Overall, the figure suggests that, on average, robots and AI play a larger role in the production structure of high-income countries than in middle-income countries. This helps explain why the simulated effects on the skill premium are stronger for the high-income group.

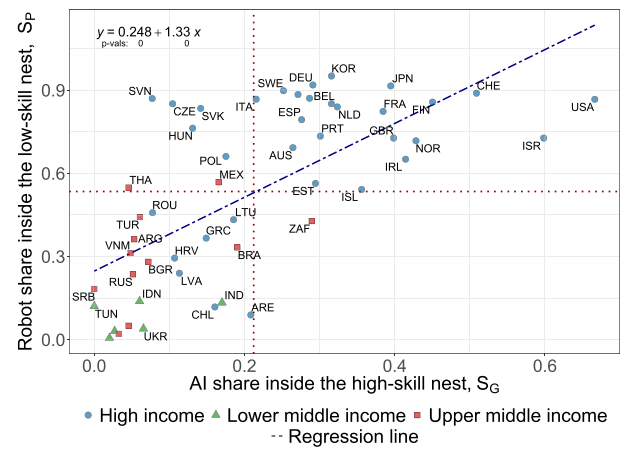


Fig. 8. Relative importance of robots in the low-skill-labor composite and AI in the high-skill-labor composite. *Notes:* (1) The vertical dotted line represents the average relative importance of AI in the high-skill-labor composite, while the horizontal dotted line represents the average relative importance of robots in the low-skill-labor composite. These two measures are defined in Eqs. (4) and (6), respectively. (2) To estimate regressions robust to outliers, we employ the Huber regressor.

Table 3

Robustness checks under random parameter draws: Effects on the skill premium variation (%).

Income level	Technology	Mean	Median	Share positive
High income	AI	−20.8	−26.2	32.9
High income	robot	77.3	38.1	67.0
High income	both	40.6	2.1	51.1
Upper m. income	AI	−13.1	−16.9	32.9
Upper m. income	robot	64.0	28.2	67.0
Upper m. income	both	42.0	8.9	55.3
Lower m. income	AI	−8.1	−12.9	32.9
Lower m. income	robot	38.9	14.9	67.0
Lower m. income	both	28.4	6.1	54.8

5.3. Robustness checks

5.3.1. Variations in the substitutability and weight parameters θ , φ , β_1 , and β_2

We conduct a robustness exercise to assess the sensitivity of our results to variations in the key parameters governing the skill-premium response. In particular, we vary the substitutability parameters θ and φ , which determine the sign and strength of the effects of robots and AI on the skill premium. We also vary the weights β_1 and β_2 , which determine the relative importance of low-skilled labor and high-skilled labor in their respective CES composites and therefore affect the magnitude of these responses. To this end, we generate 600,000 random parameter draws, assuming that θ , φ , β_1 , and β_2 follow uniform distributions between 0.01 and 0.99. We then consider three scenarios for each income group. In the first scenario, only the stock of robots increases tenfold. In the second scenario, only the stock of AI increases tenfold. Finally, in the third scenario, both technologies increase simultaneously by a factor of ten. For each scenario, we compute descriptive statistics for the percentage variation in the skill premium.¹³

The results are reported in Table 3. The first two scenarios are consistent with our previous findings. On average, an increase in the stock of robots tends to raise the skill premium, while an increase in the stock

¹³ In Appendix C.2, we show how each parameter affects the percentage variation in the skill premium when varied in isolation.

Table 4

Robustness checks under random parameter draws and alternative proxies for AI stock: Effects on the skill premium variation (%).

Income Level	Technology	Bloom et al. (2025) proxy			PIM-based proxy		
		Mean	Median	Share Positive (%)	Mean	Median	Share Positive (%)
High income	AI	−21.5	−28.8	32.9	−21.7	−28.3	32.9
High income	robot	75.8	36.5	67.0	77.3	38.0	67.0
High income	both	38.4	−1.5	49.2	39.0	0.2	50.1
Upper m. inc.	AI	−20.0	−26.0	32.9	−14.1	−20.3	32.9
Upper m. inc.	robot	61.0	26.1	67.0	63.6	27.9	67.0
Upper m. inc.	both	29.7	−4.6	47.4	40.0	7.2	54.2
Lower m. inc.	AI	−16.3	−20.1	32.9	−8.9	−15.4	32.9
Lower m. inc.	robot	36.5	13.7	67.0	39.0	15.0	67.0
Lower m. inc.	both	14.8	−7.8	44.7	28.1	5.7	54.3

of AI tends to reduce it. Furthermore, the magnitude of these effects is larger among high-income countries and the share of positive variations is consistent with the theoretical predictions.

In the third scenario, the result is determined by the interaction between two opposing forces: the robot channel, which tends to increase the skill premium, and the AI channel, which tends to reduce it. When robots and AI increase simultaneously, the net effect on the skill premium is positive on average, but it is not uniformly positive across the entire parameter space. The sign of the joint effect is approximately governed by $(\theta - \gamma)S_p - (\varphi - \gamma)S_G$. In this exercise, on average, $(\theta - \gamma)S_p > (\varphi - \gamma)S_G$. In other words, the robot channel tends to dominate on average, but this dominance depends on the relative strength of the two channels.

5.3.2. Alternative proxies for the stock of AI

Ideally, the preferred measure for AI would be the stock of AI capital per worker. However, to the best of our knowledge, no previous study has estimated this variable. We therefore construct two alternative proxies for the stock of AI per worker and perform additional robustness checks, similar to the previous exercise, to assess whether our results are sensitive to the choice of AI proxy. As in the previous robustness exercise, we assume that θ , φ , β_1 , and β_2 are uniformly distributed between 0.01 and 0.99.

To construct the first proxy for the AI stock, we follow Bloom et al. (2025) and assume that AI per worker is a linear function of robots per worker, that is, $g = 2 \times p$. To construct the second proxy for the AI stock, we follow the logic of the perpetual inventory method (PIM). Following the steady-state approach discussed by Harberger (1978) and Nehru and Dhareshwar (1993), we assume the steady-state relationship between investment in AI (I_G), the growth rate of investment in AI (g_G), and the depreciation rate of AI capital (δ_G), as implied by the Solow model. In this case, the stock of AI can be written as

$$G = \frac{I_G}{\delta_G + g_G}. \quad (11)$$

Since the exact values of δ_G and g_G are unknown, we assume that δ_G and g_G are uniformly distributed between 0.001 and 0.315, and between 0.001 and 0.26, respectively.¹⁴ For each scenario, we generate 600,000 random parameter draws and compute descriptive statistics for the percentage variation in the skill premium.

The results are shown in Table 4. Overall, the results for both proxies are consistent with our previous findings. In the AI scenario, the average

effect on the skill premium is negative for all income groups, suggesting that the AI channel continues to reduce the skill premium. In the robot scenario, the average effect is positive, as in the previous exercises. The effects are also larger in magnitude among high-income countries.

When both technologies increase simultaneously, the net effect on the skill premium remains positive, on average, under both alternative AI-stock proxies. However, the median effects and the share of positive parameter draws suggest that the AI channel plays a relatively stronger role under the Bloom et al. (2025) proxy. This is consistent with the assumption that AI per worker is larger than robots per worker in that specification. Overall, these results confirm that the combined outcome is governed by two opposing forces: the robot channel, which tends to increase the skill premium, and the AI channel, which tends to reduce it. The net effect therefore depends on the relative strength of these two channels across parameter draws and income groups.

6. Summary and concluding remarks

We build on Bloom et al. (2025) to investigate how industrial robots and AI relate to the skill premium between high-skilled and low-skilled workers in 52 countries. Using a nested CES production function, we demonstrate that the relationship between these technologies and the skill premium depends on the degree of substitutability between the technologies and human labor.

Our results indicate that an increase in the use of industrial robots is related to an increase in the skill premium because of more intensive substitution for low-skilled workers. By contrast, AI is related to a reduction in the skill premium because it substitutes for tasks performed by high-skilled workers. These results are in line with the findings of Bloom et al. (2025) and with economic intuition. Furthermore, we show that both industrial robots and AI have a positive impact on economic growth.

Fundamentally, our results represent theoretical model implications under specified parameters rather than empirical estimates. The conclusions are conditional on (1) nested CES validity, (2) applicability of externally calibrated elasticities, and (3) the assumption that the parameters are common across countries. While we performed an extensive set of robustness checks, our approach nevertheless has some shortcomings.

First, the lack of a reliable and internationally comparable proxy for the stock of AI technologies in use across countries remains an important limitation. While the correlation between investment and the stock of AI is likely to be positive, this could nevertheless be misleading and AI investment in some lower- and middle income countries may just be driven by a small number of firms. To try to overcome this limitation, we used two alternatives to represent the AI stock. The first is based on Bloom et al. (2025), where we assume that the AI capital stock per worker is equal to twice the robot stock per worker. The second is based on the perpetual inventory method, where we used a wide range of investment growth rates and depreciation rates to simulate different AI stocks. Overall, the results of these two approaches are in line with the results in the baseline version of the model. In future research it would be highly desirable to use a more precise measure of the stock of AI

¹⁴ To define the ranges for δ_G and g_G , we first assume that the minimum values of both the growth rate and the depreciation rate are 0.1%. The upper bound for the depreciation rate of AI capital is based on the depreciation rates of the computing equipment capital stock and the computer software and databases capital stock reported in the EU KLEMS dataset (Stehrer et al., 2019). The upper bound for the growth rate is based on the average annual growth rate of investment in AI between 2013 and 2023.

across countries. Relatedly, more granular data are necessary to better understand the heterogeneous effects of AI, for example, on entry-level jobs.

Second, our analysis is based on 2019 data. This could underestimate the effects of AI because the emergence of ChatGPT and other large language models at the end of 2022 is not captured. This also holds true for the rapid progress these models have made since then. We try to address this through our sensitivity analysis in which we simulate large increases in the stock of AI. However, it would be interesting to gain more insights that are truly data-based in future research.

Third, the parameter values used in our simulations are uncertain and the results are sensitive to changes in them. We carried out several robustness checks including the use of 3D plots to capture the effects of parameter changes along two dimensions to address this point. Apart from these robustness analyses, it would be insightful to obtain more precise empirical estimates of the different parameters (particularly the elasticities of substitution between different inputs) for future research.

Fourth, for tractability reasons, we classify workers into two groups—high- and low-skilled—treating medium-skilled workers as part of the low-skilled category. While this simplification prevents us from capturing job polarization dynamics highlighted in the literature, we verified that distinguishing strictly low-skilled workers from medium-skilled ones yields only minor quantitative differences, without altering our main conclusions. Nevertheless, introducing three different types of workers in a production function with three different types of capital and carrying out the corresponding simulations would definitely be a worthwhile extension allowing us to capture issues related to wage polarization. Finally, the use of a full-fledged general equilibrium framework in which the deployment of robots and AI triggers equilibrium feedback effects is beyond the scope of this paper but definitely remains an important area for future research.

In terms of policy measures, our framework suggests the following. If new technologies that substitute for different types of workers are adopted in a balanced manner, they could increase economic growth,

while they would not increase wage inequality considerably. The reason is that the inequality-enhancing effects of robots are to some extent mitigated by the inequality-reducing effects of AI on wages. This finding suggests that upper-middle and lower-middle income countries could benefit by adopting these new technologies through faster economic growth with limited increases in wage inequality. However, whether or not overall income inequality falls with the use of robots and AI also depends on the functional income distribution, where prior research has shown a reduction in the labor share due to the use of these new technologies (Eden & Gaggl, 2018; Minniti et al., 2025). Another policy lever would be education investments, which, *ceteris paribus*, decrease the skill premium and (marginally) increase incomes. Middle income countries and high-income countries alike would therefore benefit from an increase in investments in education. Overall, it is important to note that actual policy effectiveness also depends on factors not captured in our framework: adjustment costs, general equilibrium effects, skill-job mismatches, and institutional factors.

CRedit authorship contribution statement

Marcos J. Ribeiro: Writing – review & editing, Writing – original draft, Visualization, Software, Investigation, Formal analysis, Data curation, Conceptualization. **Klaus Prettner:** Writing – review & editing, Writing – original draft, Supervision, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Marcos Ribeiro reports that financial support was provided by the State of Sao Paulo Research Foundation. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Global trends in robots and AI and worker classification

A.1. Global evolution of the number of industrial robots and private investment in AI

Fig. A.1 shows the global evolution of the number of industrial robots (Panel A) and private investments in AI (Panel B). In 1993, according to the data, there were 610,925 industrial robots in use globally. Thirty years later, in 2023, this number has reached 4,822,887 units. In percentage terms, the increase was 689%. On the other hand, in 2013, private investments in AI totaled approximately 9.45 billion USD. By 2023, this value had reached approximately 119 billion USD, representing a rise of 1159%.

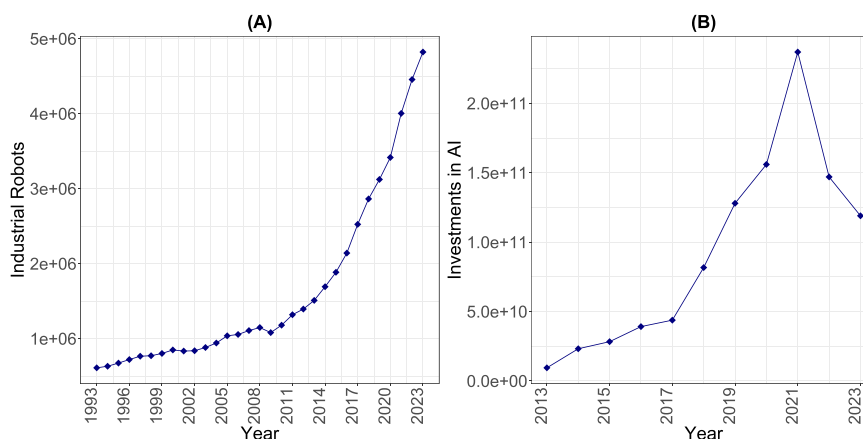


Fig. A.1. Global evolution of the number of industrial robots and private investment in AI. Note: Private investment in AI is in 2021 dollars.

A.2. Worker classification

Table A.1

International standard classification of occupations (ISCO-08) by skill level.

Skill level	Occupation
Skill levels 3 and 4 - High-skilled	Managers
	Professionals
	Technicians and associate professionals
	Clerical support workers
Skill level 1 and 2 - Low-skilled	Service and sales workers
	Skilled agricultural, forestry and fishery workers
	Craft and related trades workers
	Plant and machine operators, and assemblers
	Elementary occupations
	Armed forces occupations
Armed forces	Armed forces occupations
Not elsewhere classified	Not elsewhere classified

Notes: (1) Adapted from the International Labour Organization, see <https://ilostat.ilo.org>. (2) We do not consider workers in the armed forces and those not classified elsewhere.

Appendix B. Cross-country skill premium

Table B1 reports the countries included in our sample, their respective country codes, income classifications, and the skill premium generated by the model.

Next, we use the average hourly earnings of employees provided by the ILO to calculate the skill premium for 2019. Due to data availability, we are able to compute the observed skill premium for only 25 countries. Fig. B1 compares the skill premium generated by the model (y-axis) with the corresponding measure observed in the data (x-axis). The estimated coefficient is statistically significant only for high-income countries. Nevertheless, the relationship between the model-generated and observed skill premia is positive across all income groups.

This positive association suggests that the model is able to capture part of the cross-country variation in the observed skill premium. However, since the relationship is statistically significant only for high-income countries and the observed measure is available for a limited number of countries, this result should be interpreted as suggestive evidence rather than as a formal validation of the model.

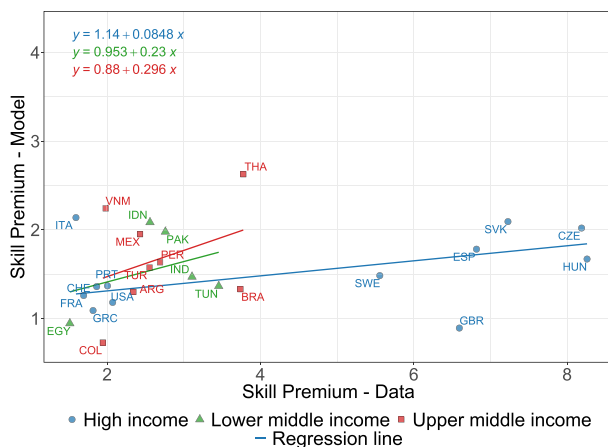


Fig. B1. Skill premium from model and data - 2019. Notes: (1) Due to a lack of data, this figure includes only 25 countries. (2) Only the coefficients of the estimated regression for high-income countries are statistically significant at the 1% level.

Table B1

Countries, codes, income classification, and the skill premium.

Country	Code	Income classification	Skill premium
Argentina	ARG	Upper middle income	1.30
Australia	AUS	High income	1.00
Austria	AUT	High income	1.79
Belgium	BEL	High income	1.45
Brazil	BRA	Upper middle income	1.33
Bulgaria	BGR	Upper middle income	1.03
Chile	CHL	High income	0.91
Colombia	COL	Upper middle income	0.73
Croatia	HRV	High income	0.85
Czech Republic	CZE	High income	2.02
Denmark	DNK	High income	1.58
Egypt	EGY	Lower middle income	0.94
Estonia	EST	High income	0.82
Finland	FIN	High income	1.29
France	FRA	High income	1.26
Germany	DEU	High income	2.07
Greece	GRC	High income	1.09
Hungary	HUN	High income	1.67
Iceland	ISL	High income	0.69
India	IND	Lower middle income	1.47
Indonesia	IDN	Lower middle income	2.08
Ireland	IRL	High income	0.91
Israel	ISR	High income	0.72
Italy	ITA	High income	2.14
Japan	JPN	High income	4.28
Latvia	LVA	High income	0.73
Lithuania	LTU	High income	0.81
Mexico	MEX	Upper middle income	1.95
Netherlands	NLD	High income	1.26
Norway	NOR	High income	0.82
Pakistan	PAK	Lower middle income	1.98
Peru	PER	Upper middle income	1.63
Poland	POL	High income	1.15
Portugal	PRT	High income	1.37
Rep. of Korea	KOR	High income	2.98
Romania	ROU	High income	1.49
Russian Federation	RUS	Upper middle income	0.68
Serbia	SRB	Upper middle income	1.05
Slovakia	SVK	High income	2.09
Slovenia	SVN	High income	1.86
South Africa	ZAF	Upper middle income	1.39
Spain	ESP	High income	1.78
Sweden	SWE	High income	1.48
Switzerland	CHE	High income	1.36
Thailand	THA	Upper middle income	2.63
Tunisia	TUN	Lower middle income	1.37
Türkiye	TUR	Upper middle income	1.57
Ukraine	UKR	Lower middle income	0.73
United Arab Emirates	ARE	High income	0.74
United Kingdom	GBR	High income	0.89
United States	USA	High income	1.18
Vietnam	VNM	Upper middle income	2.24

Appendix C. Additional results

C.1. Median levels of AI investment per worker and the stock of industrial robots

In this additional counterfactual exercise, we simulate the effects of setting AI investment per worker and the stock of industrial robots per worker equal to their respective sample median values in all countries. Since the median represents the middle value of the sample, this exercise increases AI investment per worker and the stock of industrial robots per worker in 50% of the countries. As the estimated effects of these technologies on GDP are positive, GDP increases in those countries where the counterfactual values are above the observed values, as shown in Panels A and C of Fig. C1. A similar pattern is observed for the skill premium, as shown in Panels B and D.

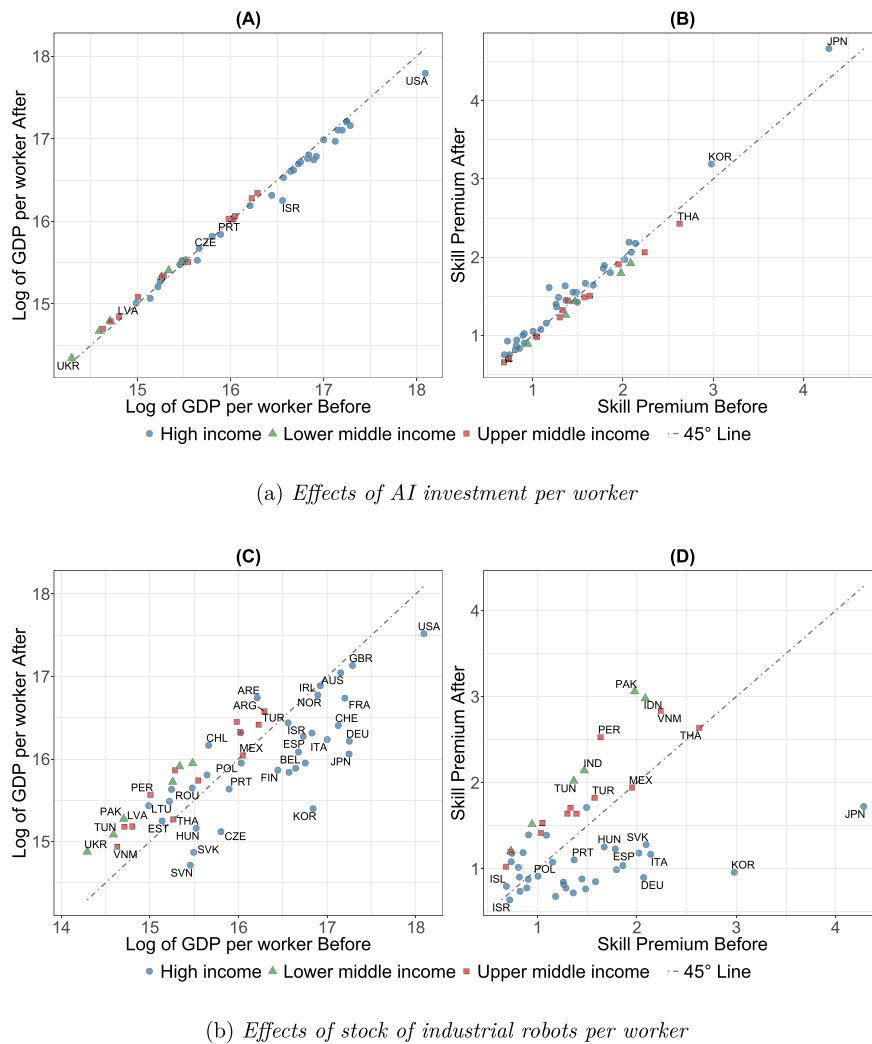


Fig. C1. GDP per worker and the skill premium before and after assigning median values of AI investment and industrial robots per worker to other countries. *Notes:* (1) In this counterfactual exercise, we assign the cross-country median values of AI investment per worker and the stock of industrial robots per worker to other countries. (2) Points above (below) the 45° line indicate that the counterfactual exercise increased (decreased) GDP per worker and the skill premium.

C.2. Changes in θ , φ , β_1 , and β_2 and the effects on skill premium

In Section 5.3, we jointly vary the substitutability parameters θ and φ together with the weights β_1 and β_2 . In this section, we vary these two groups of parameters separately to examine how each group affects the percentage change in the skill premium.

For the first group, we constructed a grid with 200,000 observations as follows: for each combination of φ and θ , ranging from 0.1 to 0.99, we assigned two pairs of multiplicative factors for robots and AI, the first equal to 1 (unchanged) and the second equal to 10. We then estimated the skill premium for each case, calculating the average, conditional on the countries' income group, and from this, the percentage change due to this increase in robots and AI.

We present the results in Fig. C2, which illustrates the relationship between the variation in the skill premium and the parameters θ , the degree of substitutability between low-skilled workers and industrial robots, and φ , the degree of substitutability between high-skilled workers and AI, for each income group: high (blue), upper-middle (green), and lower-middle (beige). In general, when observing the vertices (navy-blue points), we note that an increase in θ (with $\varphi = 0.99$) tends to intensify the variation in the skill premium across all income groups. This result is consistent with economic intuition: the better robots

replace low-skilled workers, the greater the wage gap becomes for high-skilled workers. By contrast, as φ increases (with $\theta = 0.99$), the variation in the skill premium decreases. This occurs because greater substitutability between AI and high-skilled workers reduces the relative demand for this group in the labor market, thereby putting downward pressure on the wage differential.

There is also a compensating effect between the variations of the parameters on the skill premium. For example, when both θ and φ take on high values, the result is a moderate and positive variation, stemming from the partial cancellation of opposite impacts. Robots tend to increase the skill premium, while AI tends to reduce it, although the effect of robots dominates. In the scenario in which θ is high and φ is low, the skill premium rises sharply, especially in high-income countries, highlighting the dominance of robotization. Conversely, when θ is low and φ is high, the negative effect of AI prevails.

For the second group of parameters, we did something similar to the previous group, we changed the values of β_1 and β_2 (ranging from 0.1 to 0.99). We observe in Fig. C3 that as β_1 increases (with $\beta_2 = 0.99$), the variation in the skill premium tends to decline across all groups, although it remains positive. By contrast, when β_2 increases (with $\beta_1 = 0.99$), the variation in the skill premium rises. Overall, the skill premium variation becomes negative in scenarios where β_2 is low and β_1 is high.

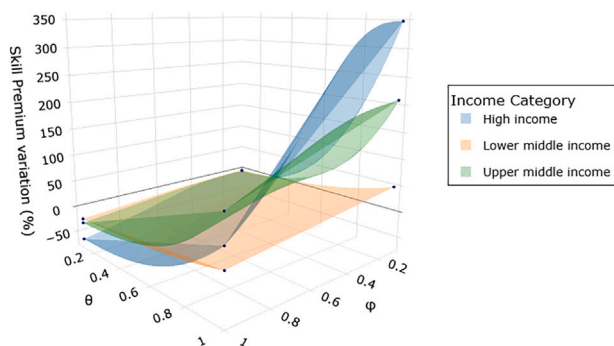


Fig. C2. Percentage changes in the skill premium due to increases in the share of robots and AI per worker, and in θ and φ . Note: The navy blue dots indicate the vertices of the planes. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

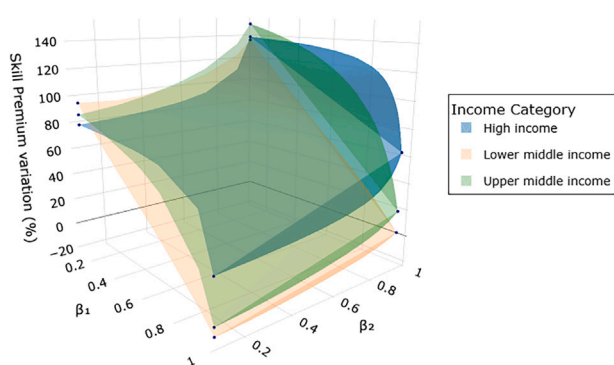


Fig. C3. Percentage changes in the skill premium due to increases in the share of robots and AI per worker, and in β_1 and β_2 . Note: The navy blue dots indicate the vertices of the planes. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Regarding β_3 , this parameter only affects the absolute magnitude of the skill premium with a higher β_3 reducing it; however, in the context of our counterfactual exercises, the percentage variation of the skill premium is independent of β_3 so we do not need to include a separate figure showing the effects of a corresponding change. All results presented in this section are consistent with the theoretical and numerical results presented earlier.

Appendix D. Supplementary data

Supplementary data for this article can be found online at doi:10.1016/j.worlddev.2026.107538.

Data availability

The replication file has been uploaded and includes all the code and data series except for the data from the International Federation of Robotics, which is proprietary and cannot be shared.

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